

# Gemini Chat

---

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

## EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [106, 158, 96, 134, 152, 129, 139, 112, 133, 153]

Tap amplitudes (pixels): [216.3631207022121, 213.92522057952868, 215.62931155109686, 219.49487465542333, 209.78083801910984, 203.8872237291979, 212.0, 217.13820483738002, 215.07673049402624, 210.27838690650069, 211.29126815843574]

- ✦ Based on the extracted features provided, here is the analysis according to the **MDS-UPDRS Item 3.4: Finger Tapping** criteria.

### Analysis of Extracted Features

#### 1. Amplitude (Decrementing):

- **Data:** The tap amplitudes range from approximately 203 to 219 pixels.
- **Trend:** The amplitudes are relatively stable. Although there is a minor dip in the middle (taps 5 and 6 drop to ~209 and ~203), the amplitude recovers immediately (taps 7 and 8 return to ~212 and ~217).
- **Conclusion:** There is no significant or persistent decrement in amplitude that characterizes the "fatigue" observed in Parkinsonism. The amplitude performance is largely normal.

#### 2. Speed and Rhythm (Interruptions/Hesitations):

- **Data:** The tap periods (frames) show significant variability, indicating an inconsistent rhythm.
  - *Fast taps:* ~96–112 frames.
  - *Slow taps (Hesitations):* There are three distinct instances where the period extends significantly,

breaking the rhythm:

- 2nd interval: **158 frames**
- 5th interval: **152 frames**
- 10th interval: **153 frames**
- **Conclusion:** The subject exhibits a "galloping" or irregular rhythm. Comparing the slow periods (approx. 155 frames) to the fast periods (approx. 100 frames), there are clear breaks in the regular tapping flow.
- **Count:** There are **3 distinct interruptions** (hesitations) in the 10-tap sequence.

### Scoring Justification

- **Normal (0):** Incorrect. The rhythm is not regular.
- **Slight (1):** Requires "one or two interruptions". The subject displays three distinct hesitations.
- **Mild (2):** Requires "**3 to 5 interruptions**". The extracted data clearly identifies 3 instances of significant slowing/hesitation (frames 158, 152, and 153) amidst faster tapping. This fits the criteria for "Mild" perfectly.
- **Moderate (3):** Requires "More than 5 interruptions" or a "freeze". The data does not show this level of severity.

The presence of 3 hesitations places this performance into the "Mild" category.